RAJAT MURHE

Final Year Computer Science Undergraduate
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PROFILE

Final year Computer Science undergraduate (BTech, AI and Machine Learning, VIT Chennai) with three internships, who works on **computer vision and video AI**: a multimodal video QA platform that checks field technician footage, and a real time object detector written in PyTorch from scratch. I build the data side as carefully as the model, with licence tracking, duplicate checks and leak free splits.

EXPERIENCE

AI Product Growth and Strategy Intern, DCS AI Technologies Sep 2026 to Present

- Integrated **third party APIs** into AI powered product workflows, connecting external services with application features and validating end to end functionality.
- Worked on a platform's multimodal AI capabilities, testing and validating image, video, audio and product ad generation workflows, including API responses and **credit consumption**.
- Contributed to an AI connector platform's integration workflows by testing connectors, dashboard operations, execution flows and **role based permissions**.
- **Debugged** integration issues and documented test cases and findings to improve product reliability and release readiness.

AI/ML Intern, Linkific | Remote Jun 2026 to Aug 2026

- Built the **backend** for a Secure Enterprise AI Assistant that answers questions from company documents, using FastAPI, RAG on PostgreSQL with pgvector and a Groq LLM.
- Added **role based access control** to the API so people only get answers from documents they are allowed to see, and helped with troubleshooting and debugging for users.

AI Intern, Rungta Ventures AI Jan 2026 to May 2026

- Built a **semantic search** feature on MyCA using embeddings, and wrote Python scripts with NLTK and spaCy that validate uploaded documents, with a team of around ten people on a live product that uses Firebase for OTP login.
- Built **RV Vision Tiny**, a real time object detector for surveillance written directly in PyTorch with no pretrained weights, with my own GIoU and BCE losses, AdamW schedule, NMS and a mAP evaluator checked against pycocotools.
- Prepared the RV Person v0.1 dataset (about 4,500 Creative Commons images) with source and licence tracking, duplicate checks and leak free splits, and designed the **annotation review** workflow. Training is not done yet.

PROJECTS

FieldProof AI, Field Service Video QA Platform | GitHub Aug 2026 to Sep 2026
Python* | *FastAPI* | *Next.js* | *TypeScript* | *PostgreSQL* | *Gemini* | *Clerk

- Built a platform where **Gemini video understanding** checks technician job videos against a checklist and a person reviews and approves each finding, with an audit trail of every job step.
- Designed a PostgreSQL job queue with retries and stale job recovery, so long video checks run on their own without the frontend open, in a multi tenant setup.

NeuroAI, Explainable Clinical Diagnostic Platform | GitHub | Live
Python* | *FastAPI* | *Scikit learn* | *XGBoost* | *PyTorch* | *BERT* | *SHAP

- Built a web app that predicts Cognitively Normal, MCI or Alzheimer's Disease from 8 patient measures. Random Forest and XGBoost scored highest at **85.23% accuracy**, with a fine tuned BERT at 81.82%, and SHAP explains each prediction.

SentinelAI, Adaptive Insider Threat Detection | GitHub | Live
Python* | *Scikit-learn* | *SHAP* | *FastAPI* | *React* | *PostgreSQL

- Built a system that spots unusual user activity that may signal an insider threat, using **Isolation Forest** and One Class SVM with Tree SHAP to explain every flag, tested with multi turn attack simulations. A research paper is in progress.

CreditX, Credit Risk Classification Model | GitHub | Live
Python* | *Pandas* | *Scikit learn* | *Streamlit

- Trained and evaluated a **classification model** on 31,711 rows, reaching 78.62% accuracy and 74.26% Macro F1, so the result is not hidden by the larger classes.

TECHNICAL SKILLS

Computer Vision and Video AI: PyTorch, object detection, GIoU and BCE losses, NMS, mAP evaluation (pycocotools), dataset preparation and annotation review, Gemini video understanding, human in the loop review

Machine Learning: Python, Scikit-learn, XGBoost, Hugging Face, BERT fine tuning, SHAP, anomaly detection, classification, Macro F1, Pandas

Backend and Deployment: FastAPI, PostgreSQL job queues, Docker, GitHub Actions CI, Git, Next.js, TypeScript, Streamlit

CS Fundamentals: Data structures and algorithms (DSA), DBMS, object oriented programming (OOPs), Python, Java, C++, Go

EDUCATION

Vellore Institute of Technology, Chennai | BTech, Computer Science, Specialization in AI and Machine Learning Aug 2023 to Sep 2027

CERTIFICATIONS AND ACHIEVEMENTS

- Won **First Prize** at the AGRHINOVA Hackathon, VIT Chennai, for a crop diagnostics platform that combines AI image classification with a multilingual RAG assistant.
- AWS Certified Developer Associate. AWS Cloud Practitioner. Google Cloud Professional ML Engineer. Microsoft Azure Developer Associate, Azure Fundamentals, ML Operations Engineer Associate. IBM Advanced Generative AI. Avery Dennison Scholarship.

LEADERSHIP AND EXTRACURRICULAR

- Vice President, Swarajya Marathi Literary Association (2025 to 2026). State Level Cricketer, Maharashtra Under 16.